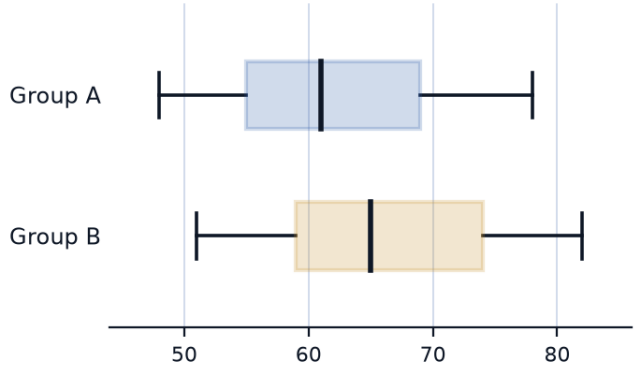
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Measures of center

Find the mean of $12, 15, 18, 9, 21$.

2 [Foundational] • Target: Measures of center

Find the median of $4, 9, 2, 11, 7, 15$.



3 [Foundational] • Target: Measures of center

Find the mode of $3, 7, 5, 7, 2, 7, 9$.

4 [Foundational] • Target: Measures of spread

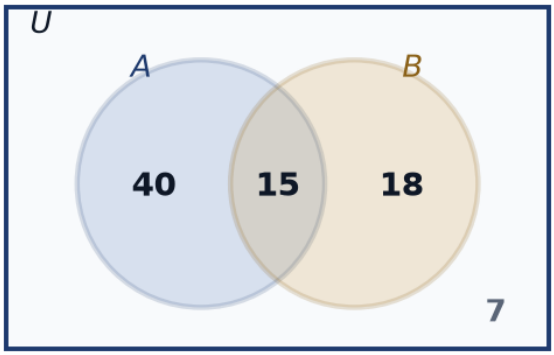
Find the range of the data set $22, 18, 35, 9, 27$.

5 [Foundational] • Target: Measures of spread

Find the interquartile range (IQR) of $4, 7, 9, 12, 15, 18, 21$.

6 [Foundational] • Target: Measures of spread

A data set has a range of 40 and a minimum value of 15. Find the maximum value.



© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Basic probability

A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Basic probability

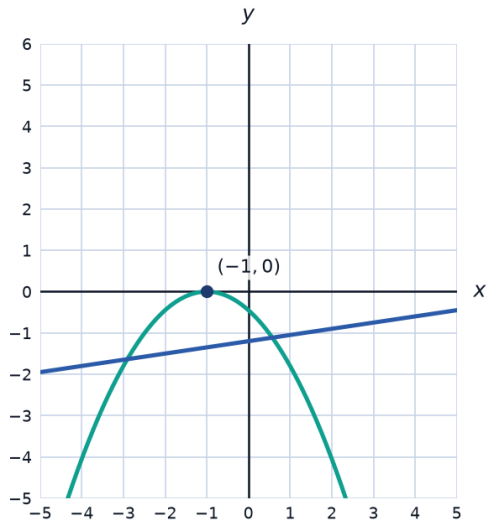
A fair six-sided die is rolled once. Find $P(\text{even number})$.

9 [Intermediate] • Target: Basic probability

A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than } 5)$.

10 [Intermediate] • Target: Combined probability: AND / OR

A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.



11 [Intermediate] • Target: Combined probability: AND / OR

Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

12 [Intermediate] • Target: Combined probability: AND / OR

A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.

13 [Intermediate] • Target: Probability without replacement

A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

14 [Intermediate] • Target: Probability without replacement

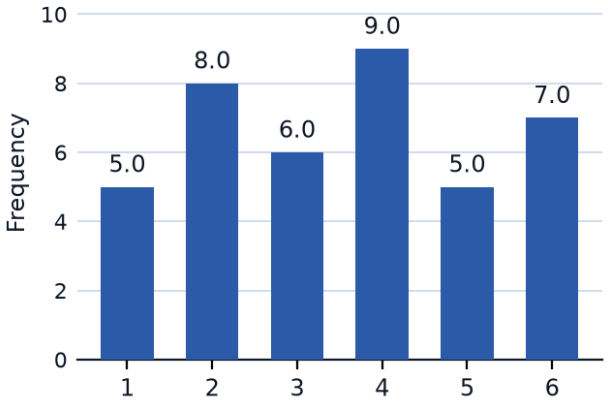
A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

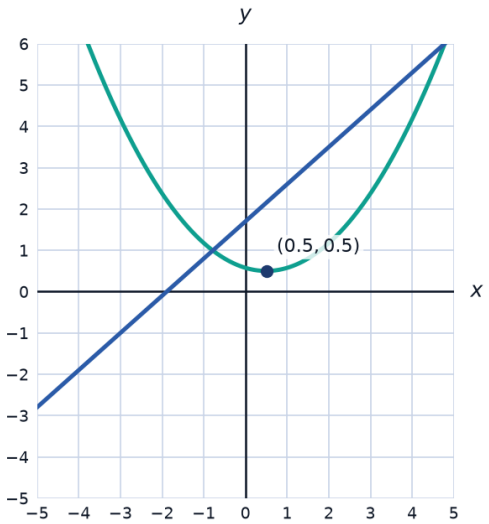
15 **[Challenging]** • **Target: Data displays** • **Cross-Lesson: 6.01 Geometry**

The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.



16 **[Challenging]** • **Target: Expected value** • **Cross-Lesson: 7.01 Trigonometry & Additional Topics**

A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.



17 **[Challenging]** • **Target: Expected value** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.

18 **[Challenging]** • **Target: Effects of transformations on statistics** • **Cross-Lesson: 1.01 Heart of Algebra**

A data set has mean 20. If every value in the set is increased by 5, find the new mean.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 2.01 Quadratics & Polynomials

A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.

20 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 3.01 Functions & Graphs

This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Measures of center

A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.

a. $\frac{6}{23}$

b. $\frac{8}{23}$

c. $\frac{11}{23}$

d. $\frac{9}{23}$

22 [Foundational] • Target: Measures of spread

A jar has 3 green, 5 blue, and 12 purple marbles. Find $P(\text{blue})$.

a. $\frac{3}{20}$

b. $\frac{7}{20}$

c. $\frac{3}{5}$

d. $\frac{1}{4}$

23 [Foundational] • Target: Basic probability

A jar has 6 red, 2 orange, and 2 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{5}$

b. $\frac{3}{5}$

c. $\frac{4}{5}$

d. $\frac{9}{10}$

24 [Foundational] • Target: Combined probability: AND / OR

A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

a. $\frac{7}{16}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Probability without replacement

A jar has 2 green, 1 orange, and 10 yellow marbles. Find $P(\text{orange})$.

a. $\frac{1}{13}$ **b.** $\frac{2}{13}$

c. $\frac{3}{13}$ **d.** $\frac{10}{13}$

26 [Foundational] • Target: Data displays

A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$ **b.** $\frac{10}{21}$

c. $\frac{11}{21}$ **d.** $\frac{3}{7}$

27 [Foundational] • Target: Expected value

A jar has 1 yellow, 5 purple, and 5 orange marbles. Find $P(\text{yellow})$.

a. $\frac{5}{11}$ **b.** $\frac{4}{11}$

c. $\frac{3}{11}$ **d.** $\frac{1}{11}$

28 [Foundational] • Target: Effects of transformations on statistics

A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

a. $\frac{1}{8}$ **b.** $\frac{1}{2}$

c. $\frac{5}{8}$ **d.** $\frac{1}{4}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a survey of students

A jar has 7 purple, 11 orange, and 10 green marbles. Find $P(\text{green})$.

a. $\frac{11}{28}$ **b.** $\frac{3}{7}$

c. $\frac{1}{4}$ **d.** $\frac{5}{14}$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Measures of center

A jar has 7 yellow, 1 green, and 9 red marbles. Find $P(\text{yellow})$.

a. $\frac{7}{17}$ **b.** $\frac{9}{17}$

c. $\frac{1}{17}$ **d.** $\frac{10}{17}$

31 [Intermediate] • Target: Measures of spread

A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{3}$ **b.** $\frac{8}{27}$

c. $\frac{10}{27}$ **d.** $\frac{4}{9}$

32 [Intermediate] • Target: Basic probability

A jar has 5 yellow, 9 orange, and 3 blue marbles. Find $P(\text{orange})$.

a. $\frac{9}{17}$ **b.** $\frac{5}{17}$

c. $\frac{11}{17}$ **d.** $\frac{3}{17}$

33 [Intermediate] • Target: Combined probability: AND / OR

A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{5}{26}$ **b.** $\frac{1}{13}$

c. $\frac{2}{13}$ **d.** $\frac{6}{13}$

34 [Intermediate] • Target: Probability without replacement

A jar has 7 red, 1 yellow, and 9 green marbles. Find $P(\text{red})$.

a. $\frac{10}{17}$ **b.** $\frac{7}{17}$

c. $\frac{9}{17}$ **d.** $\frac{1}{17}$

35 [Intermediate] • Target: Data displays

A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

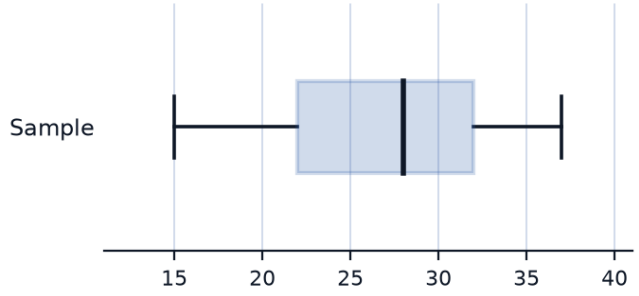
a. $\frac{5}{11}$ **b.** $\frac{9}{22}$

c. $\frac{3}{22}$ **d.** $\frac{1}{2}$

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Expected value

A data set has values 32, 22, 25, 31, 15, 37 plus one outlier, 118. By how much does the mean change (increase) when the outlier is included, compared to without it?



a. 40

b. 27

c. 14

d. 13

37 [Intermediate] • Target: Effects of transformations on statistics

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score: | Hours Studied (x) | Exam Score (y) | |---|---| | 2 | 65 | | 3 | 70 | | 4 | 78 | | 5 | 82 | | 6 | 88 | | 7 | 91 | | 8 | 95 | | 9 | 98 | | 10 | 100 | | 11 | 103 | The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 4.5x + 55$.

a. 97.5

b. 95.0

c. 100.0

d. 92.5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 1.01 Heart of Algebra

In a survey of 100 students, 60 like math, 50 like physics, and 30 like both. If a student is selected at random, what is the probability that the student likes math OR physics?

a. $\frac{4}{5}$

b. $\frac{7}{10}$

c. $\frac{3}{10}$

d. $\frac{9}{10}$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Measures of center • Cross-Lesson: 2.01 Quadratics & Polynomials

In a survey of 40 secondary students, 18 favor Math, 14 favor Science, and 6 favor both. If a secondary student is chosen at random, what is the probability that they favor Math or Science?

a. 50%

b. 55%

c. 60%

d. 65%

40 [Challenging] • Target: Measures of spread • Cross-Lesson: 3.01 Functions & Graphs

The table shows 5 students' exam scores. What is the average (arithmetic mean) of the data?

a. 82

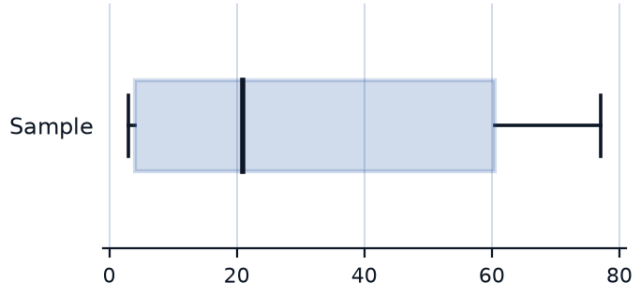
b. 84

c. 86

d. 88

41 [Challenging] • Target: Basic probability • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the range of the data set: 5, 21, 44, 77, 3?



a. 77

b. 73

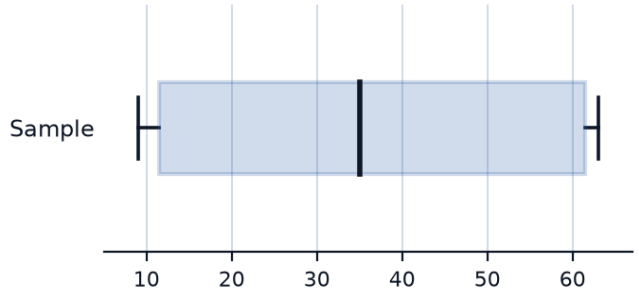
c. 74

d. 75

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Combined probability: AND / OR • Cross-Lesson: 6.01 Geometry

What is the median of the data set: 60, 63, 9, 35, 14?



a. 9

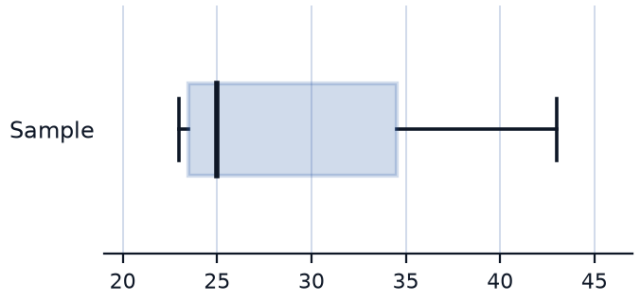
b. 35

c. 63

d. 36

43 [Challenging] • Target: Probability without replacement • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the mean (average) of the data set: 23, 24, 43, 26?



a. 28

b. 30

c. 26

d. 29

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Data displays • Cross-Lesson: 8.01 Exponentials & Logarithms

A quality control process at a manufacturing plant involves sampling widgets from two production lines, Line A and Line B. The number of defective widgets per batch of 100 is recorded. The data for the last 10 batches from each line is presented in the two-way table below. After observing this data, an engineer decides to implement a new quality improvement technique on Line B. They hypothesize that after the implementation, the mean number of defects per batch from Line B will decrease, and the variance of defects per batch will also decrease. To test this, they collect a new sample of 10 batches from Line B after the improvement. The number of defects in these new batches are: 5, 3, 6, 4, 5, 2, 7, 4, 5, 3. Initially, the production data was as follows: | Line | Batch 1 | Batch 2 | Batch 3 | Batch 4 | Batch 5 | Batch 6 | Batch 7 | Batch 8 | Batch 9 | Batch 10 | |--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------| | Line A | 8 | 7 | 9 | 8 | 10 | 7 | 8 | 9 | 8 | 9 | | Line B | 12 | 10 | 11 | 13 | 10 | 12 | 11 | 14 | 10 | 12 | Calculate the sample variance for the original Line B data and the sample variance for the new Line B data. Based on these sample variances, which of the engineer's hypotheses is directly supported by the calculated values, assuming an alpha level of 0.05 for any formal statistical test (though no formal test is required here, only comparison of sample variances)?

a. The mean number of defects decreased, and the variance decreased.

b. The mean number of defects increased, and the variance decreased.

c. The mean number of defects decreased, and the variance increased.

d. The mean number of defects increased, and the variance increased.

45 [Challenging] • Target: Expected value • Cross-Lesson: 1.01 Heart of Algebra

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students who chose Soccer or Basketball as their favorite sport?

a. 30

b. 40

c. 50

d. 60

© ClipSAT — clipsat.org | All Rights Reserved